“Smart Greenhouse & Water Tank Manager” Project

Overview

Problem: Manual watering often over/under-waters plants. Hobby greenhouses rarely track **plant comfort** or **tank level**, wasting water and harming growth.

What our prototype does. Measures **temperature & humidity (DHT11)**, **soil-moisture (ADC)**, and **tank level (HC-SR04)**. Computes **Heat Index**, **soil moisture %** (via DRY/WET calibration), and **tank fill %** (with known height). Shows results on **16×2 I²C LCD (0x27)**, indicates status with **G/Y/R LEDs**, and actuates an **SG90 servo** as a demo “valve” when soil is dry and water is available. Optional keypad sets thresholds; buzzer alerts on critical states.

Why it matters: Encourages water-efficient irrigation and healthier plants using inexpensive sensors, while exercising HAL drivers, filtering, and control logic.

Required sensors

Core Sensors and Software:

- DHT11 (Temp/RH)
- Soil-moisture sensor (analog) → ADC
- HC-SR04 ultrasonic for tank level (TRIG/echo)
- MCU: STM32F103C8 (Blue Pill), ST-LINK/V2, breadboard & jumpers.
- LCD 16×2 I²C (PCF8574 @ 0x27, 5 V) with PB6=SCL, PB7=SDA.
- 3× LEDs (G/Y/R) + resistors.
- SG90 Servo (5 V, separate supply ≥800 mA).

- Software: STM32CubeMX, Keil uVision 5, ST-Link Utility.

Functionalities

- LCD cycles: HI xx.x°C RH yy% / Soil zz% Tank ww%.
- LED status policy: Green (OK), Yellow (attention), Red (critical).
- Servo “valve” opens if **Soil < 40%** and **Tank ≥ 25%**.

Computed Metrics

- **Heat Index (°C)** from DHT11 temp/RH (Rothfusz regression).
- **Soil %** via calibrated ADC (DRY/WET).
- **Tank %** from distance with known tank height.
- **Stretch**: keypad thresholds, buzzer alarms, UART logging.

System Design

Hardware block. DHT11 (PB10), soil-moisture analog (PA0→ADC1_IN0), HC-SR04 for tank (TRIG→PA1, ECHO→PA2 via divider), LCD 16×2 I²C (PB6/PB7 @ 0x27), LEDs G/Y/R (PB0/PB1/PB2), servo SG90 (PA8 PWM). Common ground; servo on separate 5 V.

Data flow. Sensors → filtering (EMA/median) → metrics → thresholds → outputs.

Computed metrics.

- **Heat Index (°C):** from DHT11 Temp/RH (Rothfusz regression).
- **Soil moisture %:** map ADC between DRY/WET calibration points.
- **Tank fill %:** `(H - d)/H × 100` with known tank height *H* and distance *d*.

Actions & UI.

- LCD pages: `HI/Temp/RH` and `Soil%/Tank%`.
- LED policy: Green (OK), Yellow (attention), Red (critical).
- Servo "valve": open when soil < threshold and tank ≥ minimum; otherwise closed.

Firmware architecture.

- **Tasks (2–5 Hz):** read DHT/ADC/ultrasonic → compute metrics → update LED/LCD → servo control.
- **State bands:** `{OK, ATTENTION, CRITICAL}` per thresholds.
- **Calibration hooks:** store `ADC_DRY/ADC_WET`, tank height *H*.

Safety. Echo level shift; keep sensors dry; noise filtering; servo powered separately.

Design Rationale – Why We Chose This

- **Resource-efficient irrigation demo** using readily available sensors.
- **Meaningful analytics:** Heat Index, soil %, tank % → actionable decisions (servo valve).
- **Reusability:** drivers and filtering patterns carry over to many IoT/embedded builds.
- **Lab-friendly:** simple calibration steps; deterministic behavior via thresholds.

How the proposal related to our course topics

The Greenhouse systems are concrete applications of CSE331Lab concepts—exact same MCU, tools, peripherals, and design methods—showcasing the full stack from hardware wiring and driver code to real-time control and user interface.